M2 INTERNSHIP SUBJECT • 6 MONTHS

AI agents and logical reasoning for the correction of regulated metabolic models

Keywords: Systems biology, answer set programming (ASP), hybrid dynamical systems, attractors, model repair, AI agent.

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The internship can take place at IRISA (Rennes) or at LMF (Gif-sur-Yvette), at the student's choice.

Desired profile: Master's 2 in computer science, applied mathematics, or bioinformatics. No prior knowledge of biology or ASP is required; the biological context will be introduced at the beginning of the internship, with the support of the project's biologists. The internship will provide a dual expertise: answer set programming (ASP + linear programming, CEGAR approaches), and the design and evaluation of AI agent workflows (LLMs with tool access, literature search, hallucination control). Continuing into a PhD is possible. Funding is available through the ANR REGMETABO project.

Context

Bacteria are complex systems made of interconnected processes that operate at different time scales. Here, we are interested in two scales of biological processes: *metabolism* and *gene regulation*. *Metabolism* transforms nutrients from the medium into energy and biomass. Its activity is modeled by a linear program (*Flux Balance Analysis*, FBA) that computes a flux distribution maximizing biomass production. *Gene regulation* activates or inhibits reactions and gene transcription, which allows the bacterium to adapt to its environment. It is modeled by Boolean rules, *e.g.* "if regulator A is active and nutrient B is absent, then gene C is turned off ($C = \neg A \vee B$)".

Dynamics of regulated metabolism. Although strongly coupled, metabolism and regulation are most often studied separately, and models of regulated metabolism remain rare. The latest genome-scale Boolean regulatory model of *E. coli* dates from 2004 [2]. These models are simulated by *regulated FBA* (rFBA [1]), which simulates the dynamics of regulated metabolism by iterating over two steps: *(i)* the synchronous evaluation of the rules, from the state of the previous time step, determines the inhibited reactions; then *(ii)* an FBA is solved.

To reconstruct these models, the team developed MERRIN [3], which infers regulation rules from *a priori* knowledge and time series. However, MERRIN has two limitations: *(i)* it is a *de novo* method, which relearns the network from scratch; and *(ii)* the required time series are costly to produce.

Biological motivation. The latest regulatory model of *E. coli*, *iMC1010*, was built for the metabolic network *iJR904* (2003, 904 genes). We want to adapt it to the current network, *iML1515* (2017, 1,515 genes). Adapting the regulation to *iML1515* therefore requires correcting existing rules and learning new ones. Rather than relearning everything with MERRIN, we aim to identify and correct the faulty rules from data that are easier to produce: *Biolog* data. *Biolog* data provide qualitative information on the growth of a bacterial population: for a set of media and strains, we know whether or not the bacterial population grew. They describe no trajectory; only the initial conditions and the growth phenotype (growth or no growth) are known. Evaluating the compatibility of a model with these data therefore requires a new criterion.

Objectives

Preliminary work. We proposed a compatibility semantics based on *rFBA fixed points*, *i.e.* regulated metabolic states that remain unchanged after re-evaluating the regulation rules. Growth in a medium is predicted *if and only if* there exists a fixed point with growth for that medium. This semantics is implemented with MerrinASP [4], the hybrid (logic + linear) ASP solver on which MERRIN relies; it makes it possible to predict growth and to identify the rules to modify in order to change a prediction.

This program serves as the basis for a correction protocol assisted by an AI agent (an LLM equipped with

search tools), in three steps: *(i)* MerrinASP identifies the genes whose rule modification makes the model compatible with the *Biolog* data; *(ii)* the agent searches the literature for the known regulations of each candidate; *(iii)* if the articles confirm an error, the agent proposes a new rule, sourced with DOIs and verbatim citations; otherwise, the candidate is discarded. Each correction triggers a recomputation of the fixed points, which generates new candidates. On *E. coli*, this proof of concept raises the compatibility of the regulated metabolic model with 13,640 *Biolog* data points from 69.66 % to 81.51 % relative to the reference model (81 rules corrected, proprietary model `Opus 5.5`).

Objective 1: formalize the LLM × MerrinASP protocol. The current protocol is a proof of concept, with neither formalization nor robust implementation. The task will be to specify it: define the inputs, outputs, and guarantees of each step. The central challenge concerns the reliability of the literature search carried out by AI agents: *how can we prevent hallucinations and interpretation errors so that the learned rules are faithful to the literature?* As an extension, the formalization will also make it possible to identify, for each subtask, specialized *open-weight* LLMs that could replace the proprietary models currently used. This is a secondary objective aiming for a more frugal correction method and greater autonomy from proprietary models.

Objective 2: semantics of rFBA attractors. The fixed-point semantics provides guarantees only for acyclic regulatory networks. A cyclic network may admit no fixed point and converge to a cyclic attractor, *i.e.* a repeating sequence of rFBA states. *Regulatory Steady-state Analysis* (RSA) [5] computes such attractors, but only for the regulatory network alone, without coupling to metabolism. The goal is to formalize an rFBA attractor semantics, which notably requires formalizing the notion of *growth* on a cyclic attractor, and then to implement it with MerrinASP. We will then compare the correction candidates obtained with those of the fixed-point semantics.

Application to *E. coli*. The protocol will be evaluated on the adaptation of *iMC1010* to the *iML1515* network, using the 13,640 *Biolog* observations used to build *iMC1010* [2]. The corrections primarily aim for biological validity, since the experimental data are themselves subject to errors. A correction consistent with the literature could therefore degrade compatibility with the data. We will measure the number of corrected rules, the rate of compatibility with the data, the reproducibility of corrections from one run to another, and the impact of the semantics (fixed points or attractors) on the corrected rules.

References

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[3] Thuillier K. et al. MERRIN: MEtabolic regulation rule INference from time series data. *Bioinformatics* 2022.

[4] Thuillier K., Siegel A., Paulevé L. CEGAR-based approach for solving combinatorial optimization modulo quantified linear arithmetics problems. *AAAI* 2024.

[5] Marmiesse L., Peyraud R., Cottret L. FlexFlux: combining metabolic flux and regulatory network analyses. *BMC Syst. Biol.* 2015.